

Optimization of Gas Storage

P1

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P1 Project, Group MOP1-03, Mathematics-Economics

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STUDENT REPORT

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Abstract

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Preface

This project is written in the fall semester of 2026 by project group 3, consisting of six students from Mathematics-Economics at Aalborg University.

tak til vejleder
vi antager basal viden, svarende til 1. semesters elev på mat-øk
brug af ai (language improvement, coding support)
brug af tikzmaker til at lave alle tikz- og circuitkz billeder

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1 Introduction

1.1 Problem Analysis

1.2 Problem statement

2 | Financial Theory

[*This chapter is based on Capinski and Zastawniak (2021) p. 21 - 38*]
Kort intro til hvorfor dette er relevant samt en lille "teaser" til indholdet.

2.1 The Time Value of Money

The following chapter will look into *the time value of money*, and how such as interest and compounding methods link and affect the value of both present day and future day. The main idea follows that a finite payment made today is more valuable than, if same were made into the future.

This idea builds around three main issues.

- **Investment opportunity:** Money that is available now has the possibility to be invested in bonds or deposited in a bank with positive interest, making the value grow.
- **Inflation:** Money due in the future is not to be spent right away. Meanwhile, prices may rise, resulting in a change of the purchasing power of the same value, compared to the present day.
- **Risk:** Even if very small, there is always a risk, that the money is never to be received, when looking at a future payment. Having the present value reduced, to compensate for the risk.

In the following, we will be considering situations with no such risk.

2.2 Simple Interest

Simple interest is a method of calculating interest based only on the original principal not including any previously accumulated interest. It is important to remember that the principal remains unchanged during the period of the investment.

If an initial amount P is invested at a constant annual interest rate $r > 0$, the interest earned after one year is equal to rP. Therefore, the future value of the investment becomes

$$V(1) = P + rP = P(1 + r)$$

The value of the investment after t years thus becomes

$$V(t) = (1 + tr)P \quad , \quad t \in \mathbb{R}_{\geq 0} \tag{2.1}$$

Where the factor $1 + tr$ is referred to as the growth factor.

Example 2.2.1.

Suppose that 500€ is paid into a bank account where it is to earn an interest of 8% for 80 days. The future value then becomes

$$V(\frac{80}{365}) = (1 + \frac{80}{365} \cdot 0.08) \cdot 500 \approx 508.77€$$

A special application of interest appears in the valuation of a perpetuity, where a sequence of fixed payments continues indefinitely into the future. Since the annual interest rate earned on a principal P is rP, for a perpetuity paying C each year, the annual interest must satisfy

$$rP = C$$

Isolating P gives

$$P = \frac{C}{r}$$

Since the annual interest payment equals the withdrawal, the principal remains unchanged and can continue indefinitely. Although mathematically useful, simple interest is mainly of theoretical interest, as it does not count for interest being earned on previously accumulated interest. To model long-term investments more realistically compound interest must be considered.

2.3 Compound Interest

This far we have gone on the assumption, that previously accumulated interest would not be included when calculating the interest. However, we can find the compound interest, which takes the previously accumulated interest and adds it to the principal at fixed intervals. Any future interest will therefore be affected by the interest accumulated thus far.

When looking at the annual compound, it can be found by

$$V(t) = P(1 + r)^t, \tag{2.2}$$

where t is the amount of years from the starting point.

If we want to compound more than once a year, we need to introduce a new variable m, where m is the amount of compounds per year. When introducing the new variable we get that

$$V(t) = P(1 + \frac{r}{m})^{mt}, \tag{2.3}$$

which is the formula for periodic compounding.

2.3.1 Continuous Compounding

Continuous compounding is similar to periodic compounding, when the compounding frequency approaches ∞. The formula for continuous compounding can be written as:

$$V(t) = P(1 + \frac{r}{m})^{mt} = P((1 + \frac{r}{m})^{\frac{m}{r}})^{rt} \tag{2.4}$$

Theorem 2.3.1: Formula for continuous compounding

By taking the limit as $m \to \infty$ in equation 2.4, we get

$$V(t) = Pe^{rt}$$

Proof 2.1: Continuous compounding

Let $r > 0$. We note that

$$(1 + \frac{r}{m})^{\frac{m}{r}},$$

is just another version of the sequence

$$(1 + \frac{1}{n})^{n}.$$

By taking $\lim_{n\to\infty}$, we get the formula for the constant e.

$$e = \lim_{n\to\infty}(1 + \frac{1}{n})^{n}.$$

Let $n = \frac{m}{r}$. Since $r > 0$ we have $\lim_{n} \to \infty$, when $lim_{m\to\infty}$, so we get

$$e = \lim_{m\to\infty}(1 + \frac{1}{\frac{m}{r}})^{\frac{m}{r}} = \lim_{m\to\infty}(1 + \frac{r}{m})^{\frac{m}{r}}.$$

By substituting this expression into the formula for periodic compounding, we get

$$V(t) = P(\lim_{m\to\infty}(1 + \frac{r}{m})^{\frac{m}{r}})^{rt}$$

$$V(t) = Pe^{rt}.$$

This concludes the proof of the formula for continuous compounding. ■

By finding the derivative of the formula for continuous compounding we get

$$V'(t) = rPe^{tr} = rV(t)$$

This shows that the rate of growth is proportional to the current wealth. The formula for continuous compounding provides a good approximation for cases of periodic compounding, where the frequency m is large. The higher value m, the closer the case of periodic compounding is to the case of continuous compounding.

2.4 Discounting and Present Value

It is possible to convert the continuously compounded future value formula into a formula that describes the present value of a future payment.

$$V(0) = V(t) \cdot e^{-tr}$$

For this formula the discount factor is e^{-tr}.

Example 2.4.1.

An example could be that a future payment of 5000€ is received in 5 years and the rate is 5%. The question is, how much is 5000€ worth today?

$$V(0) = V(t) \cdot e^{-tr} = 5000 \cdot e^{-5 \cdot 0{,}05} \approx 3894€$$

If you invest around 3894€ and wait for 5 years, you will have 5000€ provided that the rate is 5% and the rate is yearly.

Instead of a value at time 0, you can change the formula, so it is the value at time t.

$$V(t) = e^{-r(T-t)}V(T)$$

For this formula $V(T)$ is the terminal value, and the discount factor is $e^{-r(T-t)}$.

The return for an investment for continuous compounding is not additive. So it is smart to introduce the logarithmic return.

$$k(s,t) = ln\frac{V(t)}{V(s)}$$

In this case the logarithmic return is additive, so it is possible to say.

$$k(s,t) + k(t,u) = k(s,u)$$

This theorem can be proven, but in this project it will not.

2.5 Comparing Compounding Methods

Comparing the different compounding methods is very useful when looking at which methods will produce a higher future value. The idea is that, one compounding method may be equivalent, better or worse than another method, making one of the two *preferable*. Two methods are said to be *equivalent* when the growth factors over a period of one year are the same.

Example 2.5.1.

This example will show the calculations and effective rates of four different frequency compoundings showing the effective rates over a year, with the interestrate of 8%

Comparing compounds		
	Calculation	Effective rates
Annual compound	$1 + 0,08$	$1,08$
Semi-annual compound	$\left(1 + \frac{0,08}{2}\right)^2$	$1,0816$
Monthly compound	$\left(1 + \frac{0,08}{12}\right)^{12}$	$\approx 1,083$
Daily compound	$\left(1 + \frac{0,08}{365}\right)^{365}$	$\approx 1,08328$

This example shows us, that over a period of a year, the more frequent compounds are done, the more effective they are. Even when the rate is the same across them all.

3 | Graph theory

intro og evt. definitioner:)
When optimizing a gas storage, it is fundamental to understand how graphs work. In particular, understanding directed weighted graphs and paths through graphs is essential to find the optimal route through a gas storage.

3.1 Graph terminology

kort intro

Definition 3.1.1: Sets.

A set is an unordered collection of distinct objects, called elements or members of the set. A set is said to contain its elements. We write $a \in A$ to denote that a is an element of the set A. The notation $a \notin A$ denotes that a is not an element of the set A. (Rosen, 2018, P.122)

Definition 3.1.2: Graphs.

A graph $G = (V, E)$ consists of V, a nonempty set of vertices (or nodes) and E, a set of edges. Each edge has either one or two vertices associated with it, called its endpoints. An edge is said to connect its endpoints. (Rosen, 2018, P.673)

Example 3.1.1.

Consider a road network between cities. The road network can be described as a graph $G = (V, E)$. Each city is represented by a vertex (plur.: vertices), and each road by an edge. The graph consist of the set V, which is all the vertices (cities) and the set E, which is all the edges (roads). By following the edges from one city to another, the graph can describes a route between the two cities.

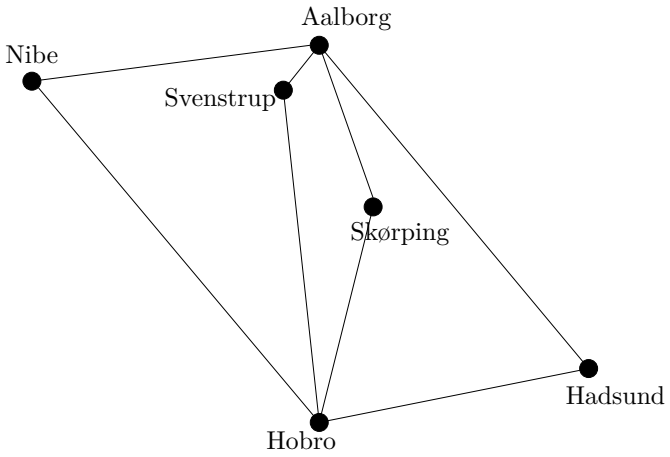


Figure 3.1: Simplified version of the road network around Aalborg.

3.1.1 Types of graphs

Different types of graphs can be characterized by the way the graph connects the *vertices* between each other with *edges*.

A graph where each edge connects two different vertices, no edge connects a vertex to itself, and no two different edges connects no same pair of vertices. A graph of such is called a **simple graph**.

Graphs that have multiple edges connecting the same pairs of vertices, are called **multigraphs**. When an unordered pair of vertices $\{u, v\}$ with m different edges connecting them, $\{u, v\}$ is said to be an edge of multiplicity m. Graphs that contain edges connecting vertices to themselves, making a *loop*, and possible multiple edges connecting the same pairs of vertices, are called **pseudographs**. The different types of undirected graphs are illustrated in figure 3.2.

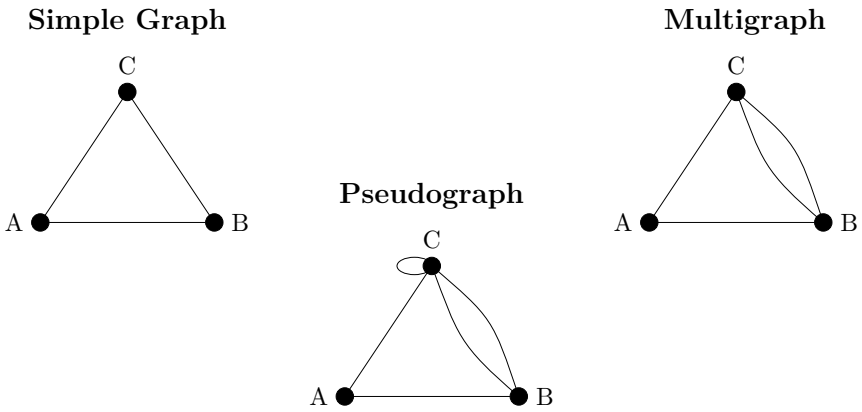


Figure 3.2: Simple graph, multigraph and pseudograph

3.1.2 Directed graphs

Instead of undirected edges, we can consider edges with a direction. This type of graph is called a directed graph. A directed graph can be useful in many situations. For example it can be used to describe a one-way road or a connection going from a starting point to an endpoint.

> **Definition 3.1.3: Directed graphs.**
>
> A directed graph (also called digraph) consists of some vertices and directed edges (also called arcs). Every directed edge goes from one vertex to another vertex, and some even goes from one vertex to the same vertex. A directed edge is denoted (u, v) is said to start in u and end in v. (Rosen, 2018, P.675)

Directed graphs and undirected graphs are lined up the same way. Some examples of directed graphs is a directed multigraph and a directed simple graph. The conditions are more or less the same. The only difference is the direction. The order of a directed edge (u, v) is important, and tells us that u is adjacent to v and vice versa v is adjacent from u, meaning that u is the initial vertex and v is the terminal- or end vertex.(Rosen, 2018, p687)
Actually there is a graph which is a mix between a directed graph and a normal graph. This type is called a mixed graph.

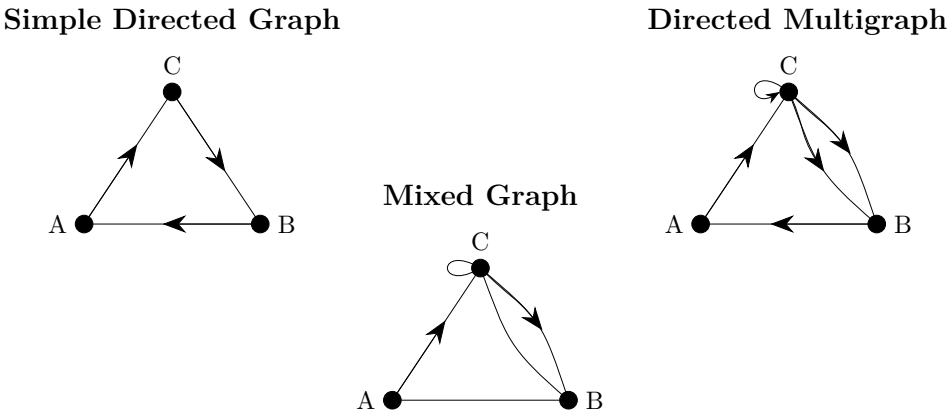


Figure 3.3: Simple directed graph, directed multigraph and mixed graph

3.2 Path

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